What factors influence the earnings of recent college graduates?

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1. Abstract

This study investigates the relationship between academic major characteristics and the median earnings of recent college graduates in the United States. Using a dataset of 172 majors, we explored variables including gender composition, employment types, and major categories. Exploratory data analysis revealed a right-skewed distribution of earnings and significant differences across fields, with Engineering yielding the highest returns. A multiple linear regression model was developed using a backwards elimination process based on the Akaike Information Criterion (AIC). The final model identified gender representation (ShareWomen) and broad major category as the primary significant predictors of income. Model diagnostics were performed to ensure adherence to linearity and homoscedasticity, and reliability was assessed through Mean Absolute Error (MAE). The final model achieved a testing MAE of $4,427, roughly 47% better than the baseline, with Engineering commanding the largest earnings premium. These findings suggest that while technical fields offer higher salaries, systemic factors like gender representation and job types continue to play a critical role in determining financial outcomes for graduates.

2. Introduction and Motivation

Choosing a college major is a high-stakes decision that can shape a student's financial trajectory for decades. As the financial stakes of higher education grow, students are increasingly evaluating their options through the lens of Return on Investment (ROI). While personal interest and passion remain important for academic success, the practical realities of debt repayment, living expenses, and long-term career prospects make understanding post-graduation earnings essential.

This project goes beyond asking "Which major pays the most?" and looks at what factors make a major more or less valuable in the job market. We examine whether gender composition, unemployment rates, and low-wage job prevalence can predict median salaries for recent graduates. The goal is to provide insights that help students choose majors wisely, guide academic advisors, and inform policymakers about labor market trends and equity.

From a data science perspective, this is a real-world problem where multiple factors interact - major category, employment type, and demographics all affect earnings. By using exploratory analysis and regression modeling, we can uncover patterns in the data and make evidence-based predictions. This type of analysis is particularly valuable because it moves beyond surface-level rankings and instead quantifies the relative contribution of each factor to graduate earnings.

Research Objectives:

- **Inference**: Understand how major characteristics relate to earnings. For example, do majors with more women earn less? How do unemployment or low-wage jobs affect salaries?
- **Prediction**: Build a model to predict median earnings based on major features like gender composition, job types, and category.

This study helps students make informed choices and highlights patterns in labor market value and equity across college majors.

Note: ChatGPT (OpenAI, 2026) was used to assist with editing and phrasing in this report.

3. Data Description

The dataset used in this project, *data_college_income*, was obtained from GitHub (Jun, n.d.) and contains information about recent college graduates and the outcomes associated with different majors. The dataset is based on survey data that tracks education and labor market outcomes in the United States.

3.1 Variable Definitions

The dataset consists of 172 observations (after cleaning), with each row representing a unique academic major.

Variable	Type	Description
Rank	Discrete numeric	Ordering of majors by median salary
Major_code	Categorical (ID)	Numeric code for each major
Major	Categorical	Name of the major
Total	Continuous	Total number of graduates
Men / Women	Continuous	Number of male/female graduates
Major_category	Categorical	16 broad fields (e.g., Engineering, Arts, Business)
ShareWomen	Continuous	The proportion of women in the major (0.0 to 1.0)
Employed	Continuous	Number of employed graduates
Full_time / Part_time	Continuous	Employment type counts
Full_time_year_round	Continuous	Full-time, year-round workers
Unemployed	Continuous	Number of unemployed graduates
Unemployment_rate	Continuous	Proportion of the workforce currently without a job
Median	Continuous	The response variable; median earnings of full-time workers
P25th / P75th	Continuous	25th and 75th percentile salaries
College_jobs	Continuous	Jobs requiring a college degree
Non_college_jobs	Continuous	Jobs not requiring a degree

Variable	Type	Description
Low_wage_jobs	Continuous	Low-wage job count

3.2 Data Cleaning and Preprocessing

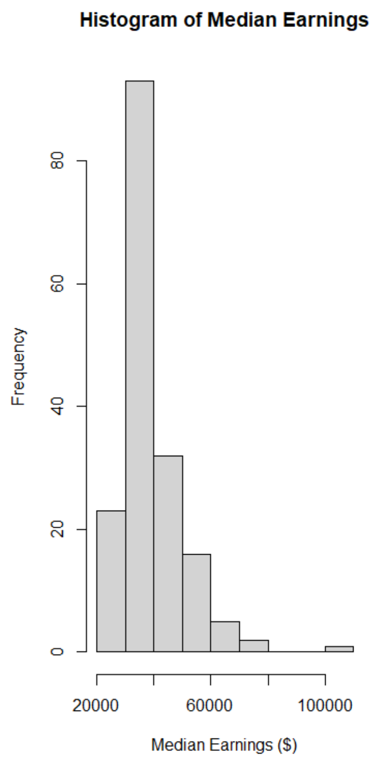
One observation corresponding to the Food Science major contained missing values in variables such as Total, Men, Women, and ShareWomen, and was removed prior to analysis.

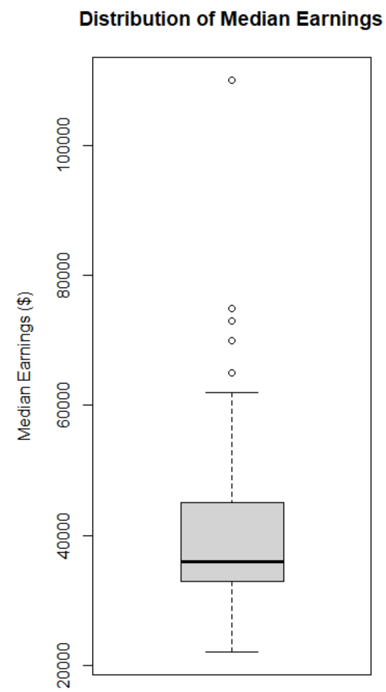
The response variable for this analysis is median earnings (Median) of full-time, year-round recent college graduates. This variable is continuous and directly reflects the income graduates earn, making it appropriate for answering the research question. Median earnings are used instead of mean earnings because the median is less sensitive to extreme values and provides a better representation of typical earnings within each major.

4. Exploratory Data Analysis

Before building a regression model, an exploratory data analysis (EDA) was conducted to understand the distribution of the response variable, examine relationships between predictors and earnings, and identify any potential issues such as skewness, outliers, or multicollinearity. Patterns identified during EDA - such as the strong negative relationship between gender composition and earnings, and the high correlation between job-type variables - directly shaped the modeling decisions made in later sections. The following subsections summarize the key findings from this process.

4.1 Distribution of the Response

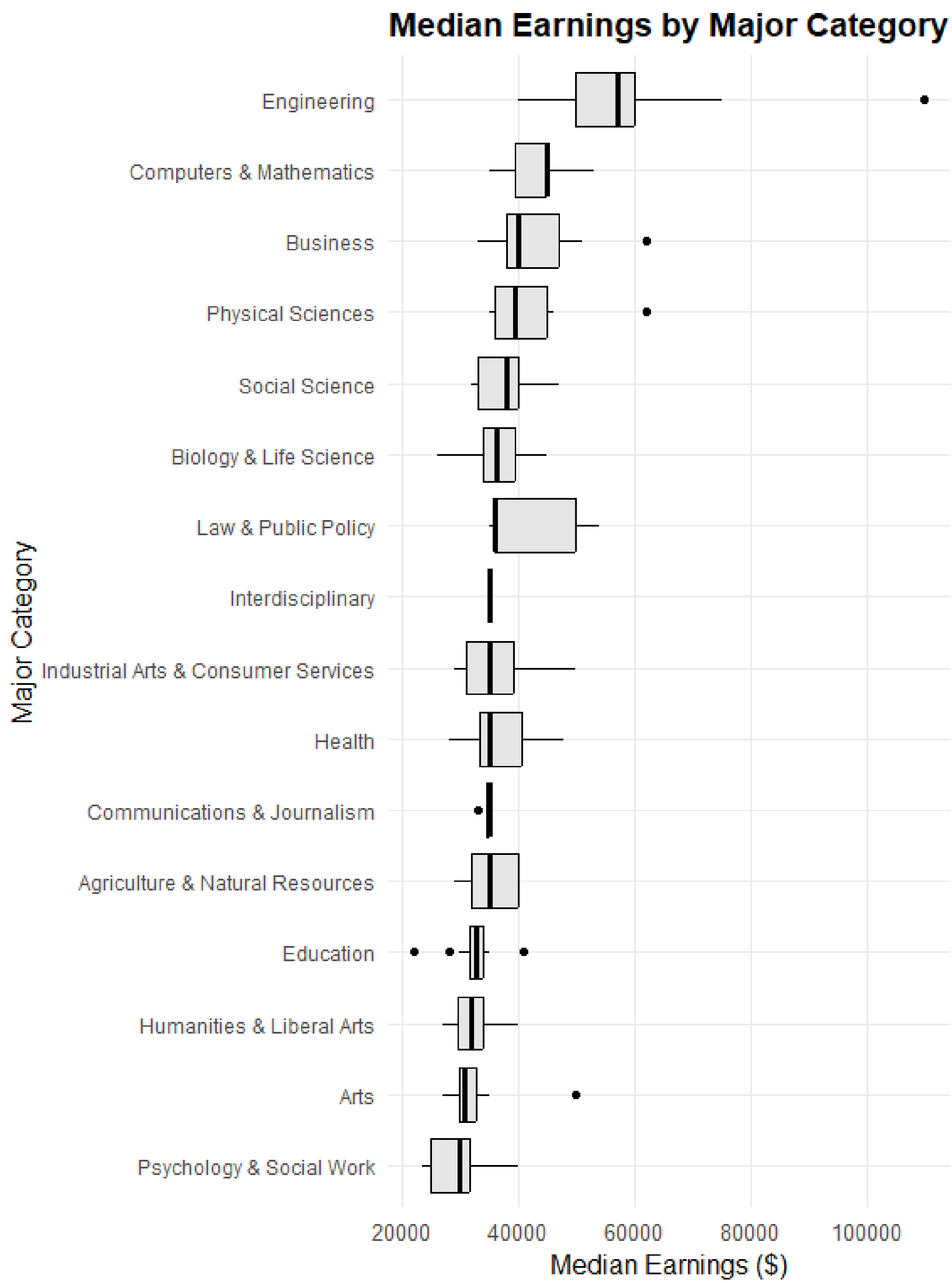




The histogram of median earnings shows that the distribution is right-skewed. Most majors have median earnings between about $30,000 and $45,000, while a smaller number of majors have much higher earnings. This suggests that although most majors have moderate salaries, a few majors earn significantly more.

The boxplot supports this observation and shows several high-value outliers, primarily in Engineering. These outliers represent majors with higher salaries, but they are still reasonable and expected, so they were not removed from the dataset.

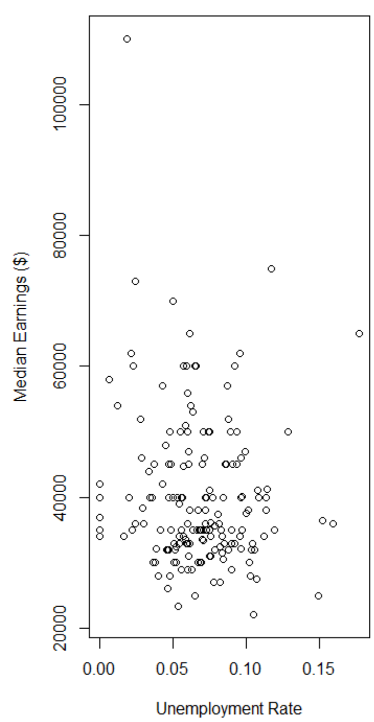
4.2 Categorical Influence



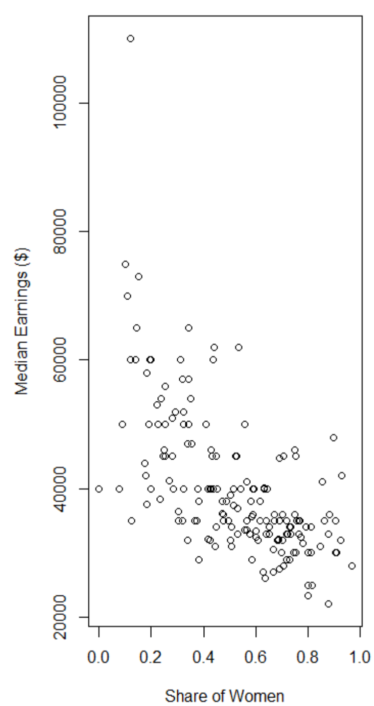
This boxplot shows clear differences in earnings across major categories. Engineering has the highest median earnings, while majors like Psychology & Social Work, Arts, and Humanities tend to have lower earnings. There are also some outliers, which means a few majors earn much more than others within the same category. Interestingly, the variance within categories like "Business" is quite high, suggesting that specific majors within that category may perform much differently than the group average.

4.3 Relationships with Predictor Variables

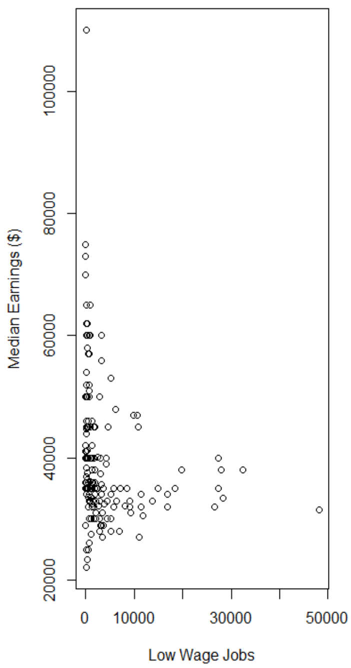
Several variables appear to be related to median earnings based on the scatterplots below. The relationship between the share of women in a major and median earnings shows a slight negative trend, where majors with a higher proportion of women tend to have somewhat lower earnings. However, the points are fairly spread out, so this relationship is not very strong.

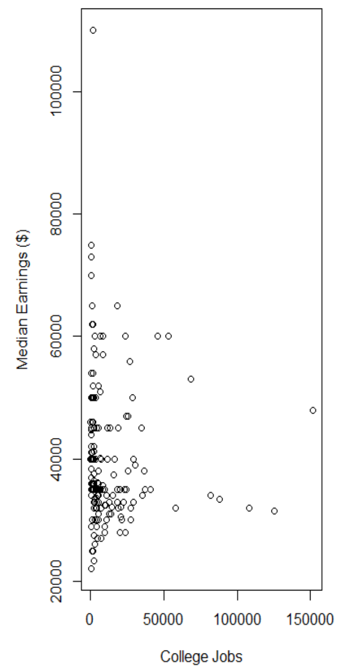


The unemployment rate does not show a strong or clear pattern with median earnings. While some majors with higher unemployment



rates tend to have lower earnings, the relationship is weak and not consistent across all data points.





The number of college jobs shows a slight positive relationship with median earnings. Majors with more graduates working in jobs that require a college degree tend to have higher earnings. However, the data is somewhat scattered, likely because this variable is influenced by how large each major is.

The number of low-wage jobs shows a clearer negative relationship with median earnings. Majors with more graduates in low-wage jobs tend to have lower earnings, which is expected and aligns with real-world patterns.

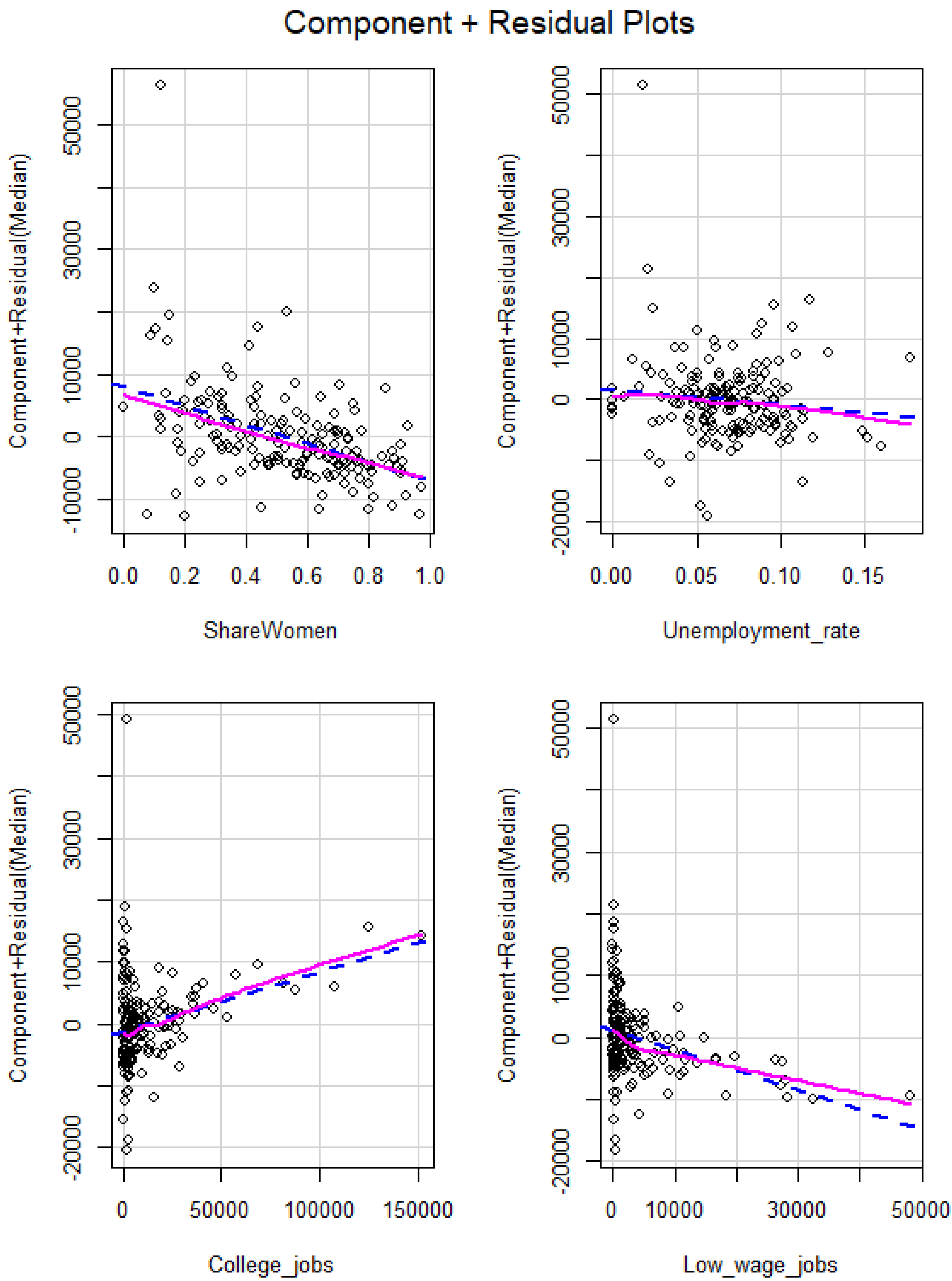
4.4 Correlation and Collinearity

```
                      Median  ShareWomen Unemployment_rate College_jobs
Median             1.00000000 -0.61868975       -0.11576905  -0.04706015
ShareWomen        -0.61868975  1.00000000        0.07320458   0.19555012
Unemployment_rate -0.11576905  0.07320458        1.00000000  -0.01169385
College_jobs      -0.04706015  0.19555012       -0.01169385   1.00000000
Non_college_jobs  -0.17181510  0.13700656        0.11956552   0.61287716
Low_wage_jobs     -0.20715284  0.18784963        0.13157271   0.64971995

                  Non_college_jobs Low_wage_jobs
Median                  -0.1718151    -0.2071528
ShareWomen               0.1370066     0.1878496
Unemployment_rate        0.1195655     0.1315727
College_jobs             0.6128772     0.6497199
Non_college_jobs         1.0000000     0.9756995
Low_wage_jobs            0.9756995     1.0000000
```

Looking at the correlation matrix above, median earnings have a moderate negative relationship with the share of women (-0.62), which suggests that majors with a higher proportion of women tend to have lower earnings. Median earnings also show weak negative relationships with unemployment rate, non-college jobs, and low-wage jobs.

Some of the predictor variables are strongly related to each other. In particular, *Non_college_jobs* and *Low_wage_jobs* are very highly correlated (0.98), and both are also strongly related to college jobs. Including all of these variables together in a model may cause issue with collinearity.

4.5 Partial Regression Plots



For the partial regression plots, the plot for share of women shows a clear negative trend, meaning that higher shares of women are associated with lower earnings. *College_jobs* show a positive relationship with earnings, while low-wage jobs show a negative relationship. The unemployment rate does not show a clear pattern, suggesting it may not be a strong predictor. As for *Major_category*, since it is a categorical variable with many levels, its relationship with median earnings was already examined using a boxplot, so a partial regression plot was not used for this variable.

5. Model Selection and Diagnostics

5.1 Data Split

To evaluate the predictive performance of this model, the 172 observations were randomly partitioned into a Training Set (70%) and a Testing Set (30%).

- The Training Set (n = 120) was used to estimate coefficients and perform variable selection.
- The Testing Set (n = 52) was used as an independent validation tool to assess how well the model generalizes to new data.

5.2 Model Selection

Based on the EDA in Section 4.4, we identified significant collinearity (r = 0.98) between *Non_college_jobs* and *Low_wage_jobs*. To prevent model instability, a Backward Elimination process was conducted using the Akaike Information Criterion (AIC). The algorithm started with a full model including gender composition, unemployment rates, and job types. Variables such as Unemployment_rate and College_jobs were removed as they did not significantly contribute to the model's predictive power. The final model retained ShareWomen and Major_category as the primary predictors of earnings.

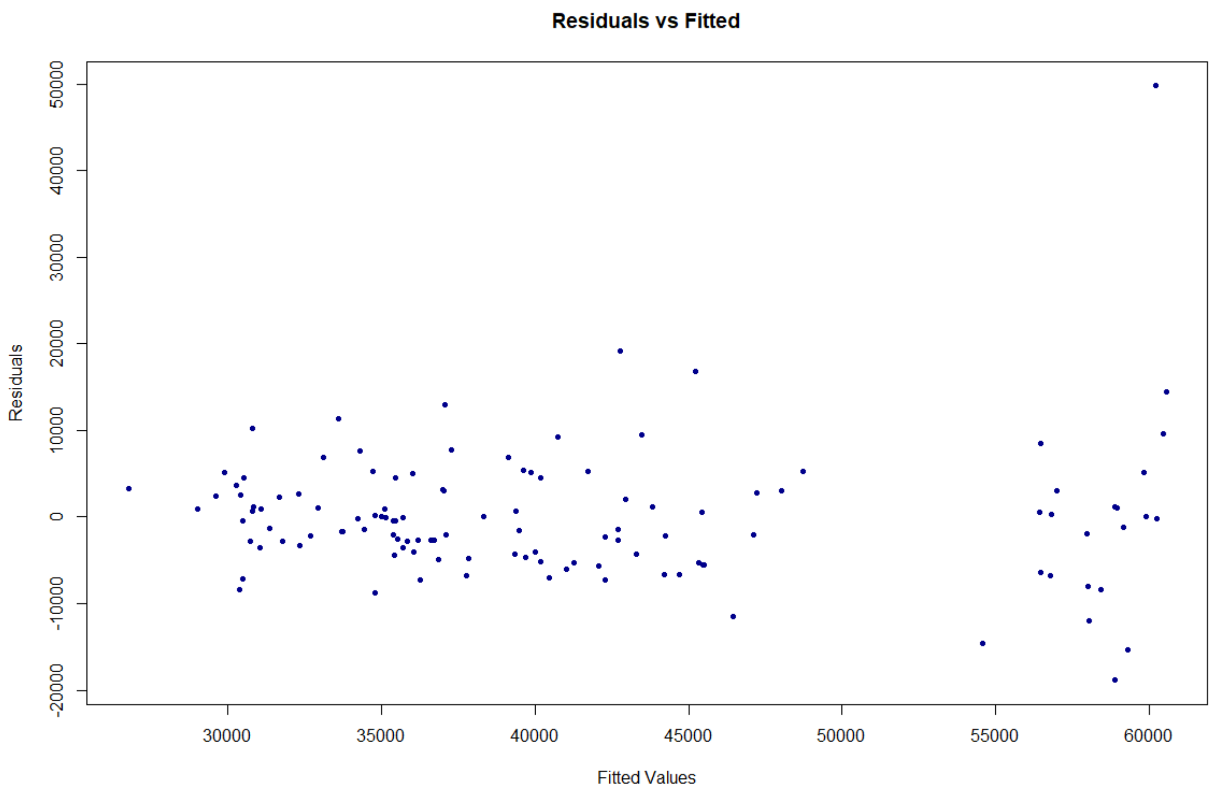
5.3 Diagnostic Testing

To ensure the statistical validity of the model, four primary diagnostic checks were performed:

- ***Multicollinearity:*** Based on the R output below, Variance Inflation Factors (VIF) were calculated for all predictors. All values remained below 4.0, with *Major_categoryEngineering* being the highest at 3.98. This confirms that the model does not suffer from significant multicollinearity.

```
Tolerance and Variance Inflation Factor
---------------------------------------
                                    Variables Tolerance      VIF
1                                  ShareWomen 0.3684698 2.713926
2                          Major_categoryArts 0.4618770 2.165078
3          Major_categoryBiology & Life Science 0.4011586 2.492779
4                      Major_categoryBusiness 0.4044704 2.472369
5      Major_categoryCommunications & Journalism 0.8226276 1.215617
6        Major_categoryComputers & Mathematics 0.4276057 2.338603
7                     Major_categoryEducation 0.3228516 3.097399
8                   Major_categoryEngineering 0.2512231 3.980526
9                        Major_categoryHealth 0.3535045 2.828818
10       Major_categoryHumanities & Liberal Arts 0.3777087 2.647543
11 Major_categoryIndustrial Arts & Consumer Services 0.4894313 2.043188
12             Major_categoryLaw & Public Policy 0.7602313 1.315389
13            Major_categoryPhysical Sciences 0.4497080 2.223665
14       Major_categoryPsychology & Social Work 0.5445267 1.836457
15               Major_categorySocial Science 0.5538643 1.805496
```

- ***Homoscedasticity***: A residuals vs. fitted plot was examined to assess the homoscedasticity assumption. The residuals appear to be roughly randomly scattered around zero across most fitted values, suggesting the assumption is largely satisfied. However, a slight increase in spread is observed at higher fitted values, likely driven by high-earning Engineering outliers identified in the Cook's Distance analysis. This mild heteroscedasticity is a common characteristic of salary data and does not severely compromise the model's validity.

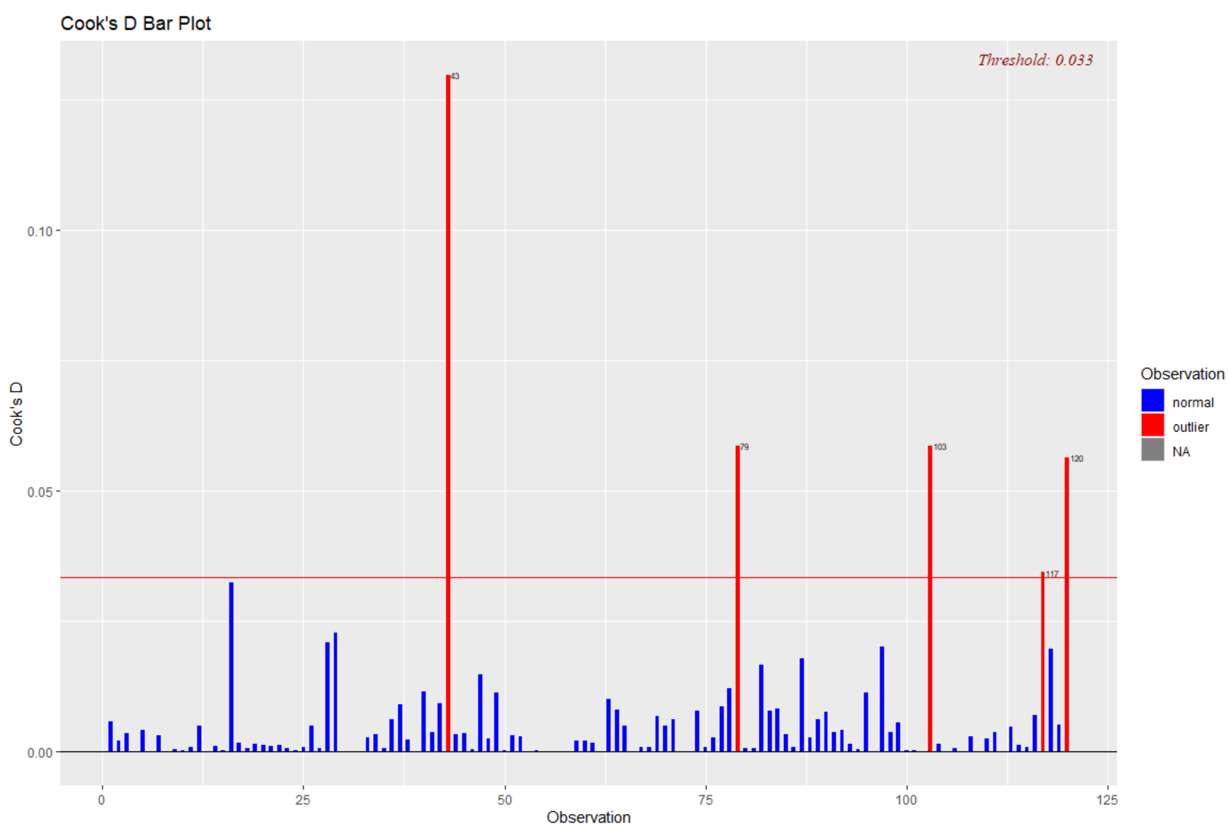


- ***Normality of Residuals:*** A Shapiro-Wilk test on the original model yielded a p-value of 0.0000000004, indicating a departure from normality. To address this, a log transformation was applied to the response variable (Median earnings). The transformed model showed a substantial improvement in normality ($W = 0.972$, $p = 0.013$). While the p-value remains below 0.05, this is likely due to the sensitivity of the Shapiro-Wilk test with moderate sample sizes - a W statistic of 0.972 is close to 1, indicating the residuals are approximately normal. While the log-transformed model improved normality (W =

0.972), the original-scale model was retained to preserve direct dollar-amount interpretability of coefficients. Both models yielded similar predictor significance.

Before Transformation	After Log Transformation
Shapiro-Wilk: W = 0.83992, p-value = 0.0000000004486	Shapiro-Wilk: W = 0.97182, p-value = 0.01271

➢ ***Influence Analysis:*** Cook's Distance was utilized to identify influential data points. Four majors (Observations 43, 79, 103, and 120) exceeded the threshold of 0.033, indicating they have a disproportionate impact on the regression line. The four influential observations were retained in the final model as they represent legitimate high-earning Engineering majors rather than data entry errors, and removing them would reduce the representativeness of the dataset.



6. Results and Model Validation

6.1 Predictive Accuracy

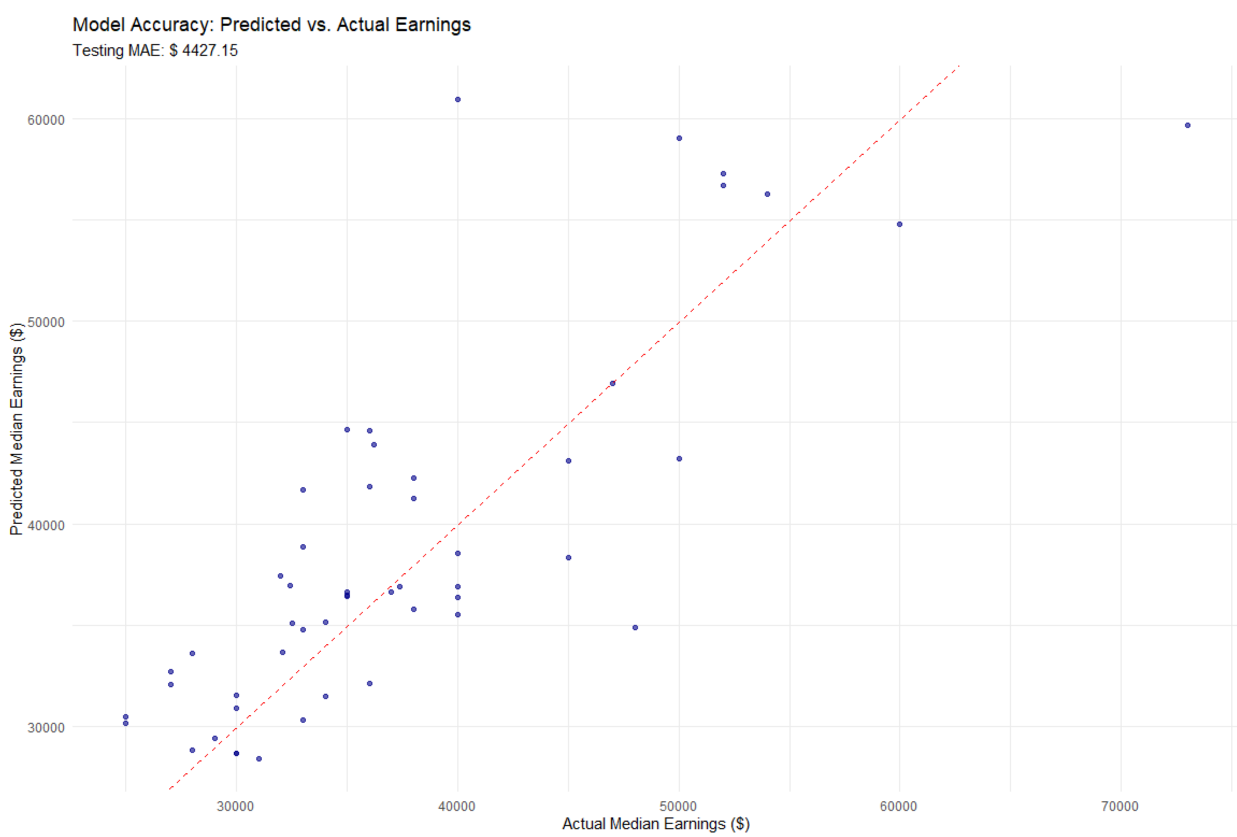
The model's performance was validated using a 70/30 data split to ensure it could generalize to unseen data:

```
Call:
lm(formula = paste(response, "~", paste(preds, collapse = " + ")),
    data = l)

Residuals:
   Min     1Q Median     3Q    Max
-18866  -4313   -430   2996  49781

Coefficients:
                                                  Estimate Std. Error t value
Pr(>|t|)
(Intercept)                                        41842.6     3909.0  10.704 <
0.0000000000000002 ***
ShareWomen                                        -17039.7     5298.5  -3.216
0.00173 **
Major_categoryArts                                  2220.1     4694.7   0.473
0.63728
Major_categoryBiology & Life Science                3821.1     4482.5   0.852
0.39592
Major_categoryBusiness                             10920.7     4254.2   2.567
0.01168 *
Major_categoryCommunications & Journalism           6074.5     9069.6   0.670
0.50449
Major_categoryComputers & Mathematics               5436.0     4341.7   1.252
0.21336
Major_categoryEducation                             3516.5     4560.9   0.771
0.44245
Major_categoryEngineering                          20431.3     3855.7   5.299
0.000000656 ***
Major_categoryHealth                                8282.8     4775.1   1.735
0.08577 .
Major_categoryHumanities & Liberal Arts              957.9     4402.4   0.218
0.82817
Major_categoryIndustrial Arts & Consumer Services    456.0     4560.7   0.100
0.92054
Major_categoryLaw & Public Policy                  10895.3     6699.3   1.626
0.10691
Major_categoryPhysical Sciences                    10069.4     4470.4   2.252
0.02639 *
Major_categoryPsychology & Social Work              2251.3     5645.4   0.399
0.69086
Major_categorySocial Science                        5938.5     5028.4   1.181
0.24030
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 8192 on 104 degrees of freedom
Multiple R-squared:  0.6026,    Adjusted R-squared:  0.5453 F-statistic: 10.51 on
15 and 104 DF,  p-value: 0.000000000000006949
```

- ***Testing MAE:*** The Mean Absolute Error on the testing set was $4,427.15, meaning the model's salary predictions are typically within this range of the actual value:

Metric	Value
Testing MAE	$4,427.15
Baseline (Mean) MAE	$8,399.58
Improvement over Baseline	~47%

- ***Explanatory Power:*** The final model explains 55% of the variance in median graduate earnings across the 172 majors analyzed. This means that major category and gender composition alone account for more than half of the differences in earnings between majors, which is a strong result given the complexity of labor market outcomes.



6.2 Final Model

The final regression model selected through backward AIC elimination is:

Median = 41,843 − 17,040 (ShareWomen) + 2,220 (Arts) + 3,821 (Biology & Life Science) + 10,921 (Business) + 6,075 (Communications & Journalism) + 5,436 (Computers & Mathematics) + 3,517 (Education) + 20,431 (Engineering) + 8,283 (Health) + 958 (Humanities & Liberal Arts) + 456 (Industrial Arts & Consumer Services) + 10,895 (Law & Public Policy) + 10,069 (Physical Sciences) + 2,251 (Psychology & Social Work) + 5,939 (Social Science) + ε

where the baseline (intercept) category is **Agriculture & Natural Resources**, and each *Major_category* coefficient represents the estimated difference in median earnings compared to that baseline group. Full coefficient estimates and significance levels are shown in the R output in Section 6.1.

6.3 Variable Impact

The academic field a student chooses is the strongest predictor of their future earnings. For example, Engineering graduates are predicted to earn about **$20,400 more per year** than graduates from Agriculture & Natural Resources majors, even when comparing students with similar gender compositions in their major. Business and Physical Sciences graduates also earn meaningfully more than the baseline, at roughly $10,900 and $10,100 more respectively.

Gender composition also plays a significant role. Majors where women make up a larger share of graduates tend to pay less - for every 10 percentage points more women in a major, the predicted median salary is about **$1,700 lower**. This holds true even when comparing majors within the same broad field, suggesting that the gender makeup of a profession itself is associated with how it is compensated in the labor market.

7. Conclusion

This research demonstrates that a significant portion of graduate earnings can be predicted by broad field categories and demographic shifts within those fields. While job types and unemployment rates were not significant factors, the "gender mix" of a major is mathematically visible in salary outcomes from day one of graduation. These results can help students, academic advisors, and policymakers make more informed decisions about the economic value of different fields of study.

7.1 Limitations and Improvements

The dataset represents a snapshot in time; recent economic shifts (e.g., the rise of AI or changes in the gig economy) may not be fully captured.

The model uses median values for entire majors, which hides the vast variation in individual success within those majors.

Additionally, the model only considers broad major categories rather than individual majors, which may oversimplify the relationship between field of study and earnings.

7.2 Reflections

This project deepened my understanding of how multiple linear regression works in a real-world context. Working with salary data showed that model assumptions like normality and homoscedasticity are rarely perfectly met in practice - the focus should be on diagnosing problems and making justified decisions rather than expecting perfect results. I became more comfortable using R tools like the olsrr package for diagnostics and rsample for data splitting. From a subject matter perspective, the finding that gender composition is a statistically significant predictor of earnings - even after controlling for field of study - was striking and reinforced the importance of thinking critically about what patterns in data actually reflect about the real world.

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